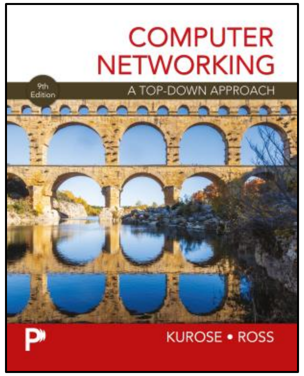


CSC 361
Computer Networks
Application Layer
(Chapter 2)

Wenjun Yang

Summer 2026

Application Layer: Overview (1 of 7)

- Principles of network applications
- Web and HTTP
- E-mail, SMTP, IMAP
- The Domain Name System DNS
- video streaming and content distribution networks
- socket programming with UDP and TCP

Application Layer: Overview (2 of 7)

Our goals:

- conceptual **and** implementation aspects of application-layer protocols
  - transport-layer service models
  - client-server paradigm
  - peer-to-peer paradigm
- learn about protocols by examining popular application-layer protocols and infrastructure
  - HTTP
  - SMTP, IMAP
  - DNS
  - video streaming systems, CDNs
- programming network applications
  - socket API

Some Network Apps

- social media
- Web
- text messaging
- e-mail
- multi-user network games
- streaming stored video (YouTube, Hulu, Netflix)
- P2P file sharing
- voice over IP
- real-time video conferencing (e.g., Zoom)
- Internet search
- remote login
- …

Q: your favorites?

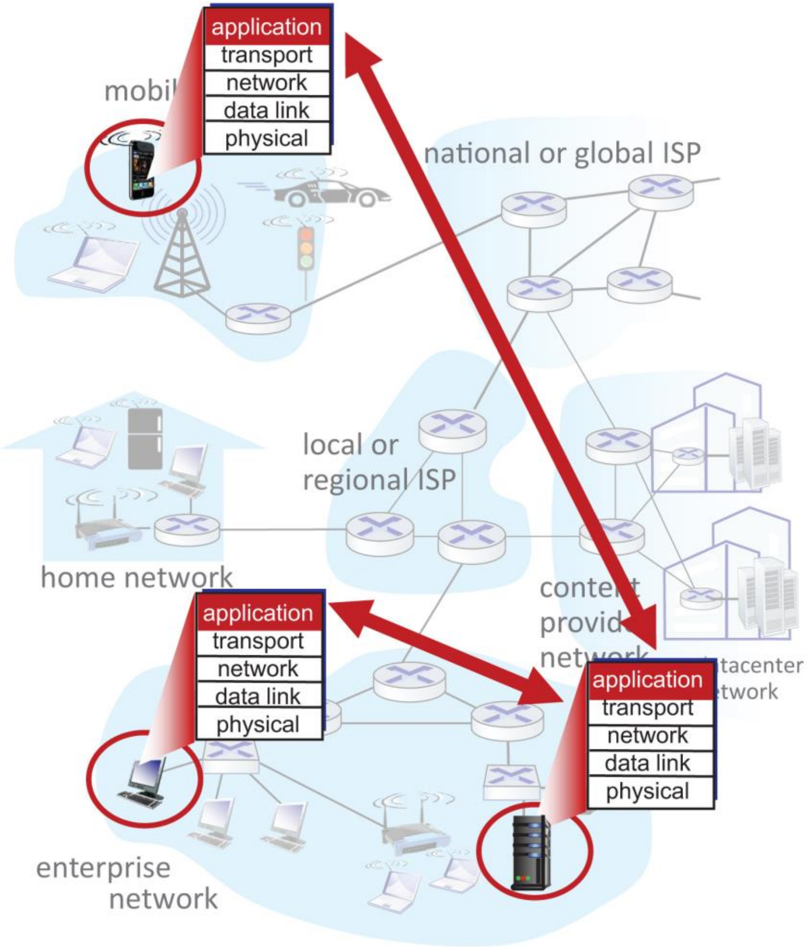
Creating a Network App

write programs that:

- run on (different) end systems
- communicate over network
- e.g., web server software communicates with browser software

no need to write software for network-core devices

- network-core devices do not run user applications
- applications on end systems allows for rapid app development, propagation



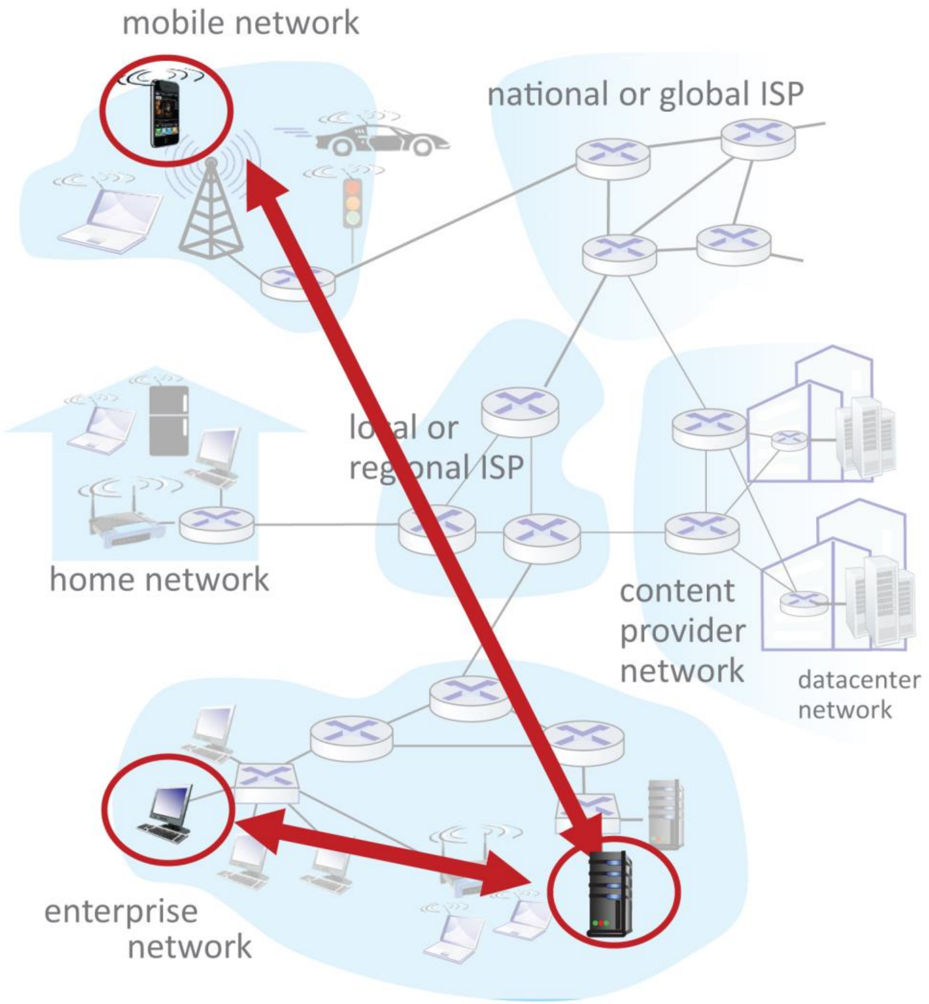
Client-Server Paradigm

server:

- always-on host
- permanent IP address
- often in data centers, for scaling

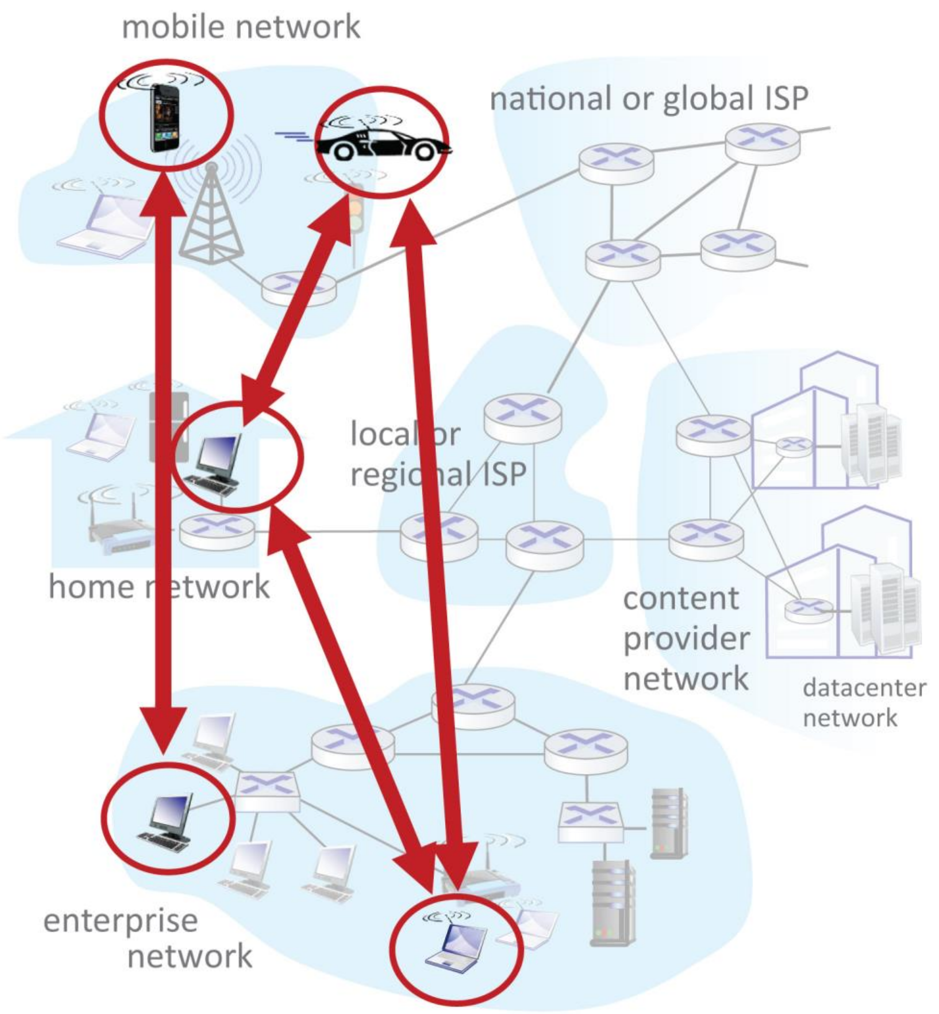
clients:

- contact, communicate with server
- may be intermittently connected
- may have dynamic IP addresses
- do **not** communicate directly with each other
- examples: HTTP, IMAP, FTP



Peer-Peer Architecture

- **no** always-on server
- arbitrary end systems directly communicate
- peers request service from other peers, provide service in return to other peers
  - **self scalability** – new peers bring new service capacity, as well as new service demands
- peers are intermittently connected and change IP addresses
  - complex management
- example: P2P file sharing [BitTorrent]



Processes Communicating

process: program running within a host

- within same host, two processes communicate using inter-process communication (defined by OS)
- processes in different hosts communicate by exchanging messages

clients, servers

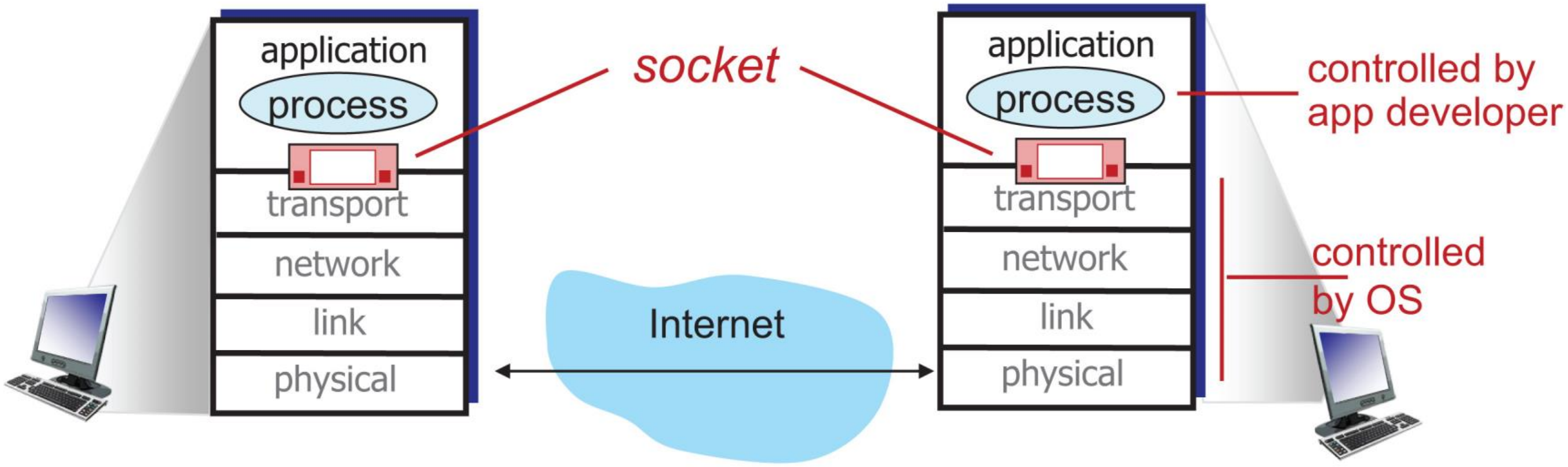
client process: process that initiates communication

server process: process that waits to be contacted

- note: applications with P2P architectures have client processes & server processes

Sockets

- process sends/receives messages to/from its socket
- socket analogous to door
  - sending process shoves message out door
  - sending process relies on transport infrastructure on other side of door to deliver message to socket at receiving process
  - two sockets involved: one on each side



Addressing Processes

- to receive messages, process must have **identifier**
- host device has unique 32-bit IP address
- *Q:* does IP address of host on which process runs suffice for identifying the process?
  - *A:* no, **many** processes can be running on same host

- **identifier** includes both IP address and port numbers associated with process on host.
- example port numbers:
  - HTTP server: 80
  - mail server: 25
- to send HTTP message to gaia.cs.umass.edu web server:
  - IP address: 128.119.245.12
  - port number: 80

An Application-Layer Protocol Defines:

- types of messages exchanged,
  - e.g., request, response
- message syntax:
  - what fields in messages & how fields are delineated
- message semantics
  - meaning of information in fields
- rules for when and how processes send & respond to messages

open protocols:

- defined in RFCs, everyone has access to protocol definition
- allows for interoperability
- e.g., HTTP, SMTP

proprietary protocols:

- e.g., Zoom

What Transport Service Does an App Need?

data integrity

- some apps (e.g., file transfer, web transactions) require 100% reliable data transfer
- other apps (e.g., audio) can tolerate some loss

timing

- some apps (e.g., Internet telephony, interactive games) require low delay to be "effective"

throughput

- some apps (e.g., multimedia) require minimum amount of throughput to be "effective"
- other apps ("elastic apps") make use of whatever throughput they get

security

- encryption, data integrity, ...

Transport Service Requirements: Common Apps

application	data loss	throughput	time sensitive?
file transfer/download	no loss	elastic	no
e-mail	no loss	elastic	no
Web documents	no loss	elastic	no
real-time audio/video	loss-tolerant	audio: 5Kbps-1Mbps video:10Kbps-5Mbps	yes, 10's msec
streaming audio/video	loss-tolerant	same as above	yes, few secs
interactive games	loss-tolerant	Kbps+	yes, 10's msec
text messaging	no loss	elastic	yes and no

Internet Transport Protocols Services

TCP service:

- **reliable transport** between sending and receiving process
- **flow control:** sender won't overwhelm receiver
- **congestion control:** throttle sender when network overloaded
- **connection-oriented:** setup required between client and server processes
- **does not provide:** timing, minimum throughput guarantee, security

UDP service:

- **unreliable data transfer** between sending and receiving process
- **does not provide:** reliability, flow control, congestion control, timing, throughput guarantee, security, or connection setup.

Q: why bother? **Why** is there a UDP?

Internet Applications, and Transport Protocols

application	application layer protocol	transport protocol
file transfer/download	FTP [RFC 959]	TCP
e-mail	SMTP [RFC 5321]	TCP
Web documents	HTTP [RFC 7230, 9110]	TCP
Internet telephony	SIP [RFC 3261], RTP [RFC 3550], or proprietary	TCP or UDP
streaming audio/video	HTTP [RFC 7230], DASH	TCP
interactive games	WOW, FPS (proprietary)	UDP or TCP

Securing TCP

Vanilla TCP & UDP sockets:

- no encryption
- cleartext passwords sent into socket traverse Internet in cleartext (!)

Transport Layer Security (TLS)

- provides encrypted TCP connections
- data integrity
- end-point authentication

TLS implemented in application layer

- apps use TLS libraries, that use TCP in turn
- cleartext sent into "socket" traverse Internet **encrypted**
- more: Chapter 8

Application Layer: Overview (3 of 7)

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Web and HTTP

First, a quick review…

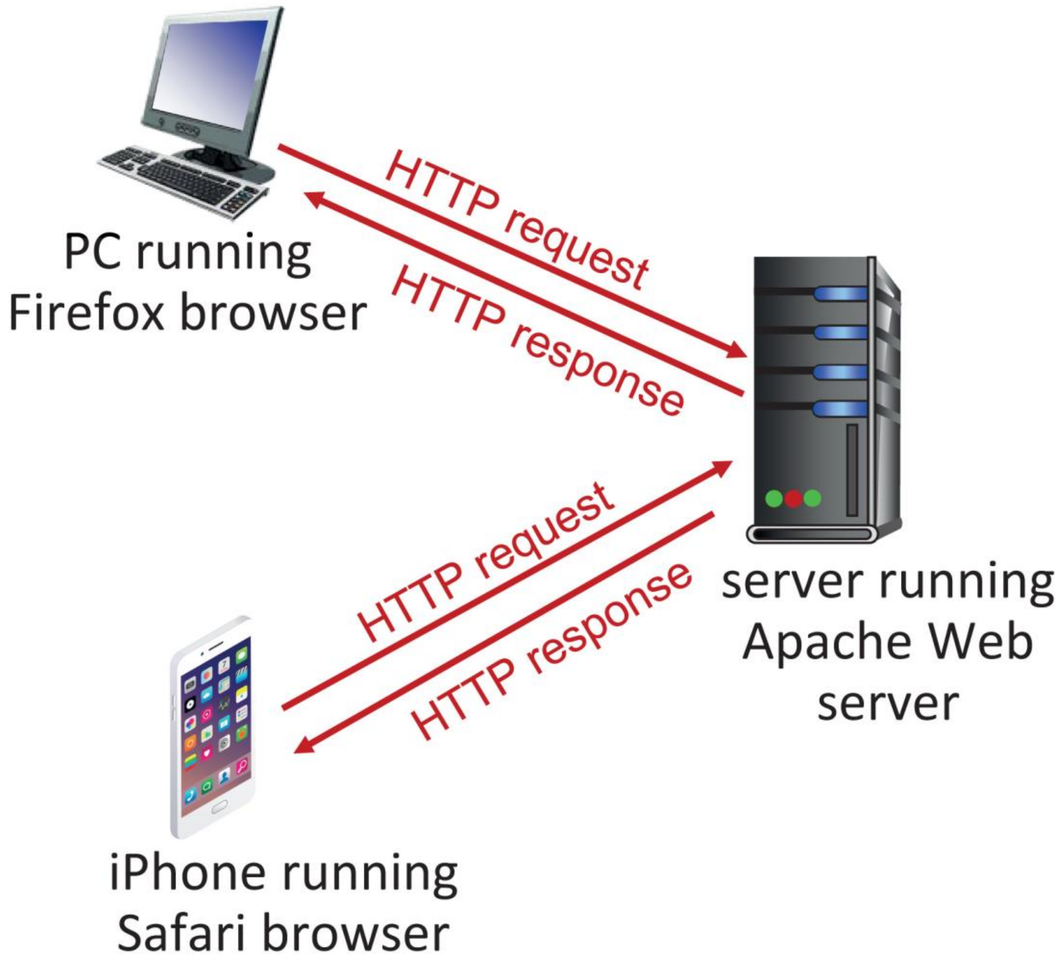
- web page consists of **objects,** each of which can be stored on different Web servers
- object can be HTML file, JPEG image, Java applet, audio file,…
- web page consists of **base HTML-file** which includes **several referenced objects, each** addressable by a **URL,** e.g.,

$$\underbrace{\texttt{www.someschool.edu}}_{\text{host name}}\underbrace{\texttt{/someDept/pic.gif}}_{\text{path name}}$$

HTTP Overview (1 of 2)

HTTP: hypertext transfer protocol

- Web's application-layer protocol
- client/server model:
  - **client:** browser that requests, receives, (using HTTP protocol) and "displays" Web objects
  - **server:** Web server sends (using HTTP protocol) objects in response to requests



HTTP Overview (2 of 2)

HTTP uses TCP:

- client initiates TCP connection (creates socket) to server, port 80
- server accepts TCP connection from client
- HTTP messages (application-layer protocol messages) exchanged between browser (HTTP client) and Web server (HTTP server)
- TCP connection closed

HTTP is "stateless"

- server maintains **no** information about past client requests

aside

protocols that maintain "state" are complex!

- past history (state) must be maintained
- if server/client crashes, their views of "state" may be inconsistent, must be reconciled

HTTP Connections: Two Types

Non-persistent HTTP

1. TCP connection opened
2. at most one object sent over TCP connection
3. TCP connection closed

downloading multiple objects required multiple connections

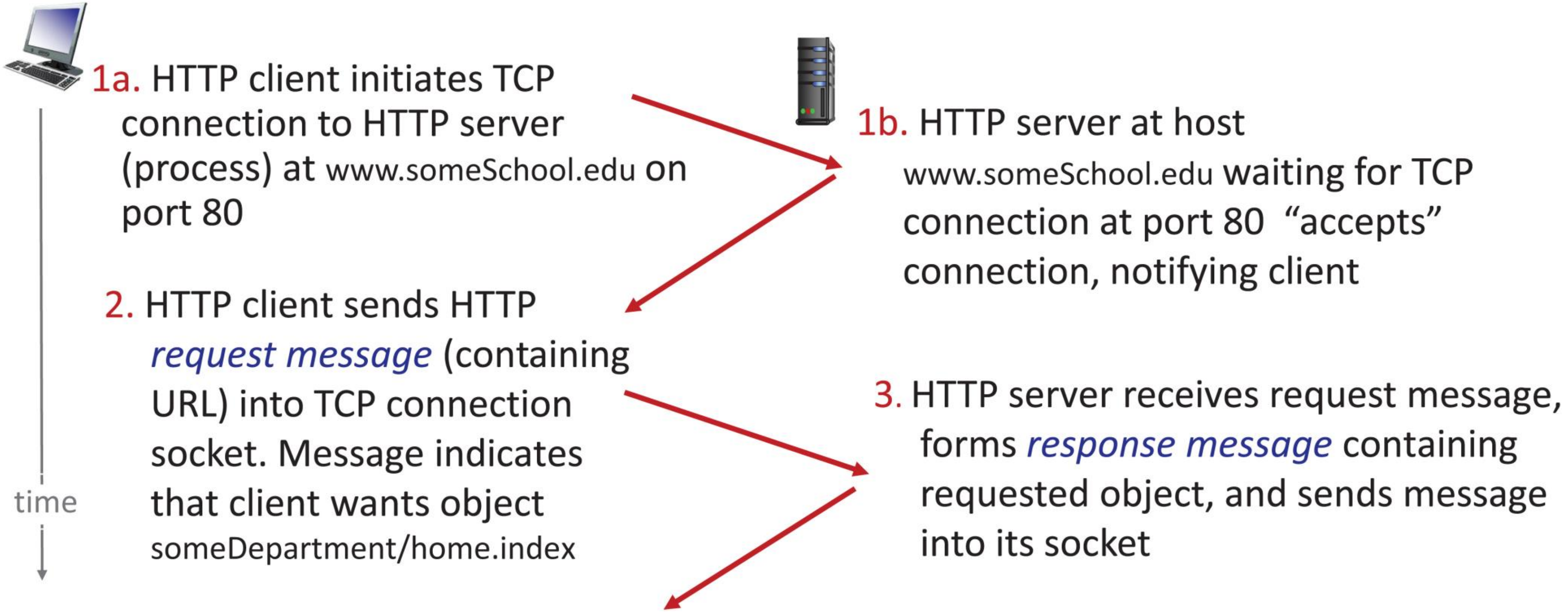
Persistent HTTP

- TCP connection opened to a server
- multiple objects can be sent over **single** TCP connection between client, and that server
- TCP connection closed

Non-Persistent HTTP: Example (1 of 2)

User enters URL: www.someSchool.edu/someDepartment/home.index

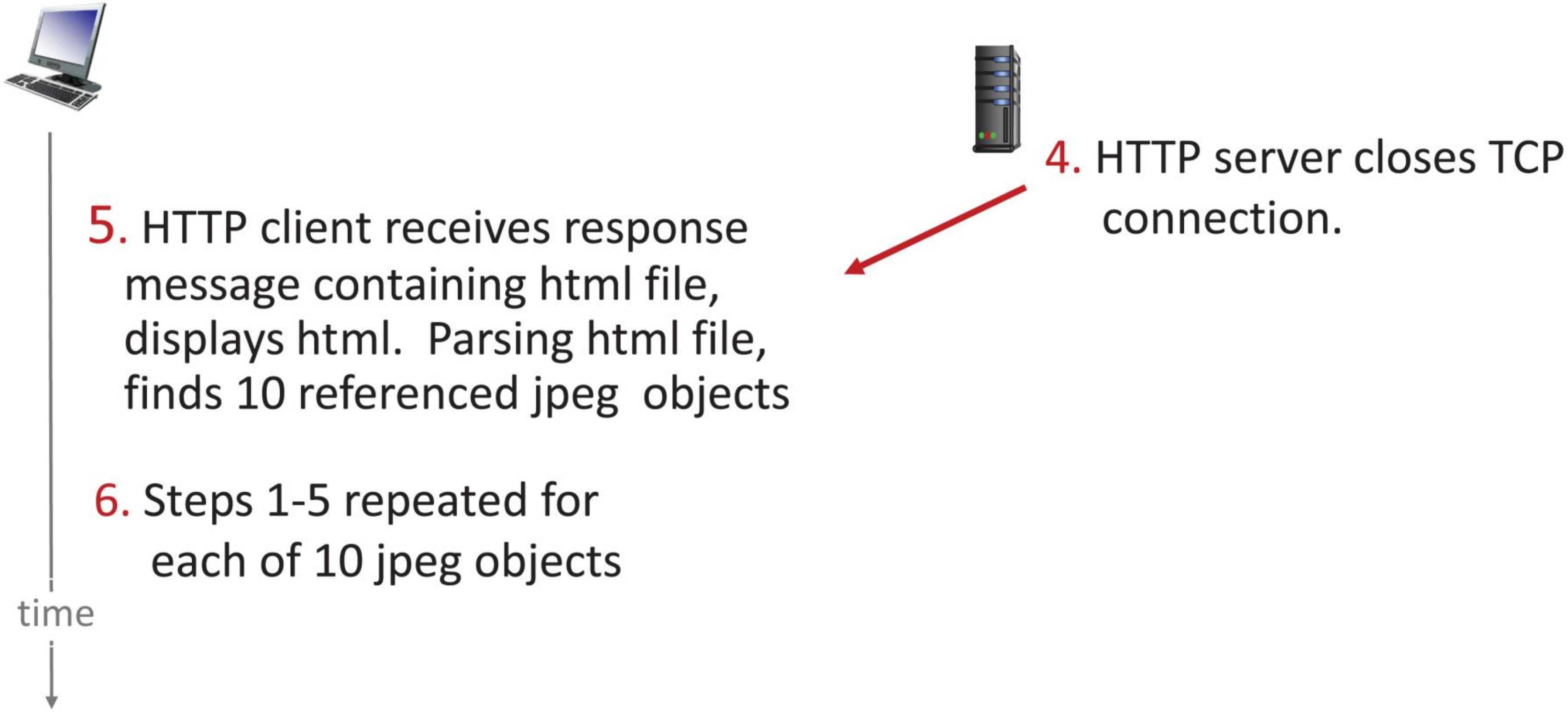
(containing text, references to 10 jpeg images)



Non-Persistent HTTP: Example (2 of 2)

User enters URL: www.someSchool.edu/someDepartment/home.index

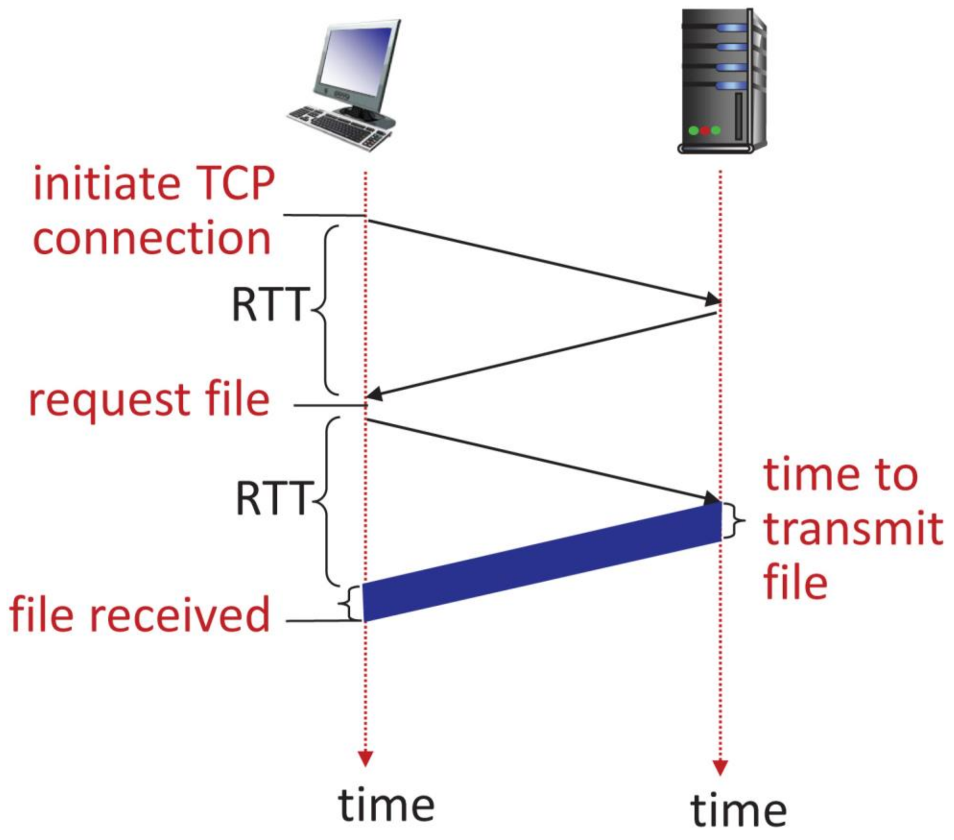
(containing text, references to 10 jpeg images)



Non-Persistent HTTP: Response Time

RTT (definition): time for a small packet to travel from client to server and back

HTTP response time (per object):
- one RTT to initiate TCP connection
- one RTT for HTTP request and first few bytes of HTTP response to return
- object/file transmission time



Non-persistent HTTP response time = 2RTT+ file transmission time

Persistent HTTP (HTTP 1.1)

Non-persistent HTTP issues:

- requires 2 RTTs per object
- OS overhead for **each** TCP connection
- browsers often open multiple parallel TCP connections to fetch referenced objects in parallel

Persistent HTTP (HTTP1.1):

- server leaves connection open after sending response
- subsequent HTTP messages between same client/server sent over open connection
- client sends requests as soon as it encounters a referenced object
- as little as one RTT for all the referenced objects (cutting response time in half)

HTTP Request Message

- two types of HTTP messages: **request, response**
- HTTP request message:
  - ASCII (human-readable format)

carriage return character

line-feed character

request line (GET, POST, HEAD commands)

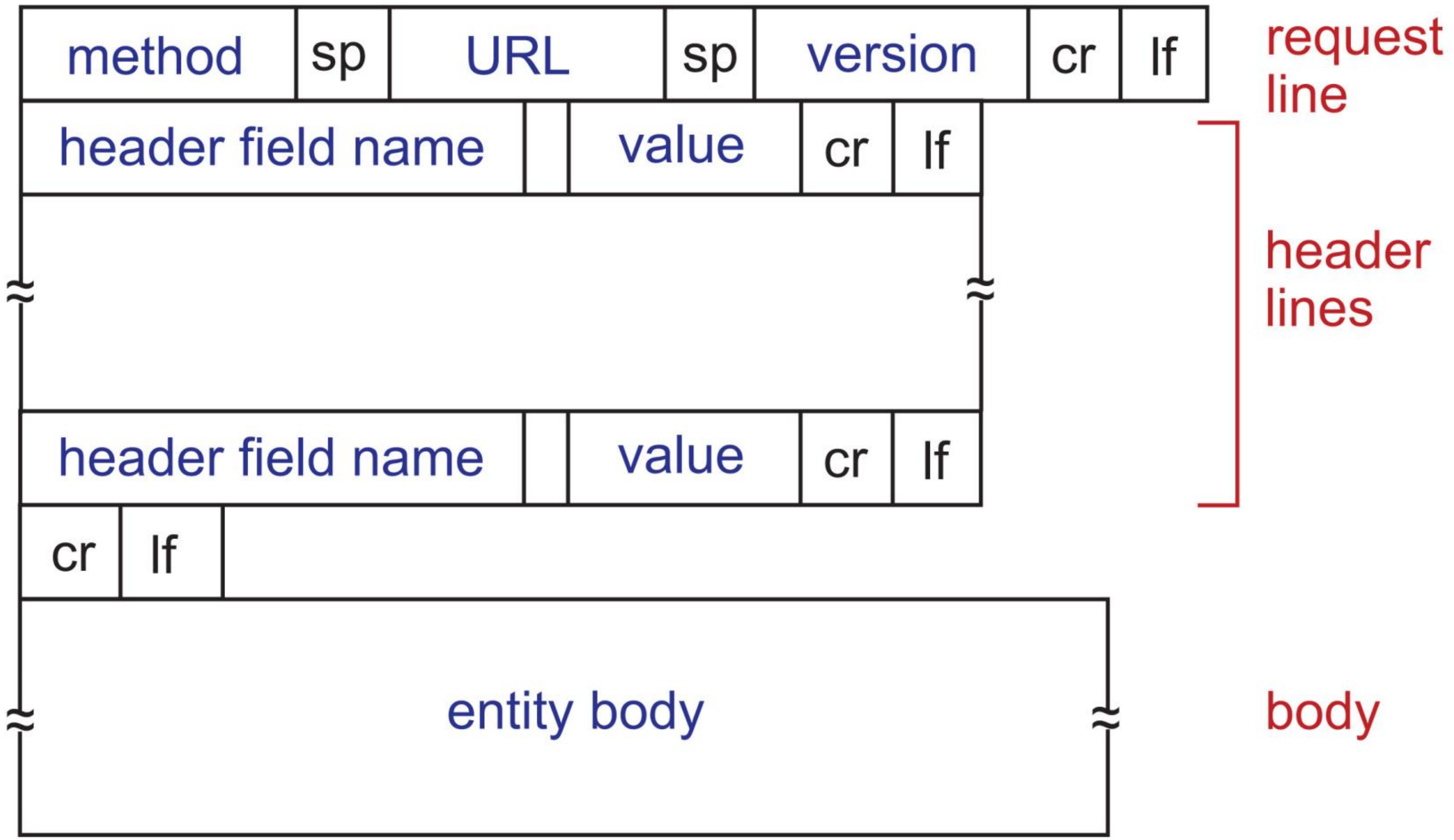
header lines

carriage return, line feed at start of line indicates end of header lines

```
GET /index.html HTTP/1.1\r\n
Host: www-net.cs.umass.edu\r\n
User-Agent: Mozilla/5.0 (Macintosh; Intel Mac OS X
   10.15; rv:80.0) Gecko/20100101 Firefox/80.0 \r\n
Accept: text/html,application/xhtml+xml\r\n
Accept-Language: en-us,en;q=0.5\r\n
Accept-Encoding: gzip,deflate\r\n
Connection: keep-alive\r\n
\r\n
```

* Check out the online interactive exercises for more examples: http://gaia.cs.umass.edu/kurose_ross/interactive/

HTTP Request Message: General Format



Other HTTP Request Messages

POST method:

- web page often includes form input
- user input sent from client to server in entity body of HTTP POST request message

GET method (for sending data to server):

- include user data in URL field of HTTP GET request message (following a '?'):

www.somesite.com/animalsearch?monkeys&banana

HEAD method:

- requests headers (only) that would be returned **if** specified URL were requested with an HTTP GET method.

PUT method:

- uploads new file (object) to server
- completely replaces file that exists at specified URL with content in entity body of POST HTTP request message

HTTP Response Message

status line (protocol status code status phrase)

header lines

data, e.g., requested HTML file

```
HTTP/1.1 200 OK
Date: Tue, 08 Sep 2020 00:53:20 GMT
Server: Apache/2.4.6 (CentOS)
   OpenSSL/1.0.2k-fips PHP/7.4.9
   mod_perl/2.0.11 Perl/v5.16.3
Last-Modified: Tue, 01 Mar 2016 18:57:50 GMT
ETag: "a5b-52d015789ee9e"
Accept-Ranges: bytes
Content-Length: 2651
Content-Type: text/html; charset=UTF-8
\r\n
data data data data data ...
```

* Check out the online interactive exercises for more examples:

http://gaia.cs.umass.edu/kurose_ross/interactive/

HTTP Response Status Codes

- status code appears in 1st line in server-to-client response message.
- some sample codes:

200 OK
- request succeeded, requested object later in this message

301 Moved Permanently
- requested object moved, new location specified later in this message (in Location: field)

400 Bad Request
- request msg not understood by server

404 Not Found
- requested document not found on this server

505 HTTP Version Not Supported

Trying out HTTP (Client Side) for Yourself

1. netcat to your favorite Web server:

% nc -c -v gaia.cs.umass.edu 80 (for Mac)

>ncat –C gaia.cs.umass.edu 80 (for Windows)

- opens TCP connection to port 80 (default HTTP server port) at gaia.cs.umass.edu
- anything typed in will be sent to port 80 at gaia.cs.umass.edu

2. type in a GET HTTP request:

```
GET/kurose_ross/interactive/index.php HTTP/1.1
Host: gaia.cs.umass.edu
```

- by typing this in (hit carriage return twice), you send this minimal (but complete) GET request to HTTP server

3. look at response message sent by HTTP server!

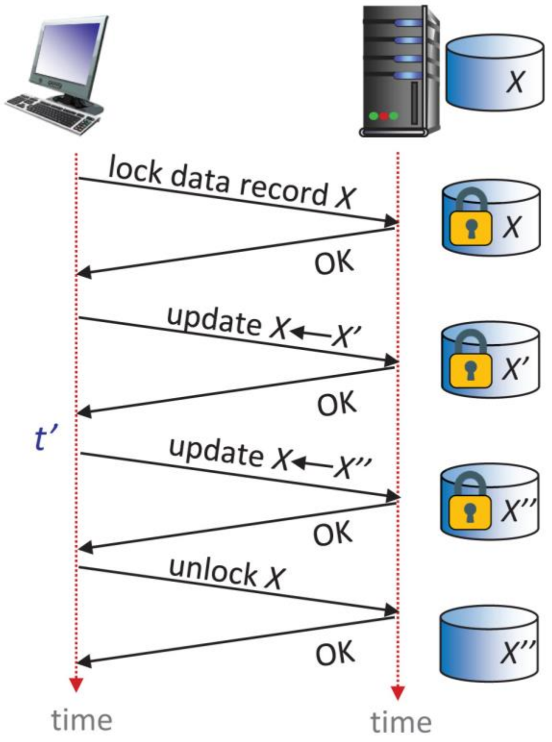
(or use Wireshark to look at captured HTTP request/response)

Maintaining User/Server State: Cookies (1 of 3)

Recall: HTTP GET/response interaction is stateless

- no notion of multi-step exchanges of HTTP messages to complete a Web "transaction"
  - no need for client/server to track "state" of multi-step exchange
  - all HTTP requests are independent of each other
  - no need for client/server to "recover" from a partially-completed-but-never-completely-completed transaction

a stateful protocol: client makes two changes to X, or none at all



Q: what happens if network connection or client crashes at *t'*?

Maintaining User/Server State: Cookies (2 of 3)

Web sites and client browser use **cookies** to maintain some state between transactions

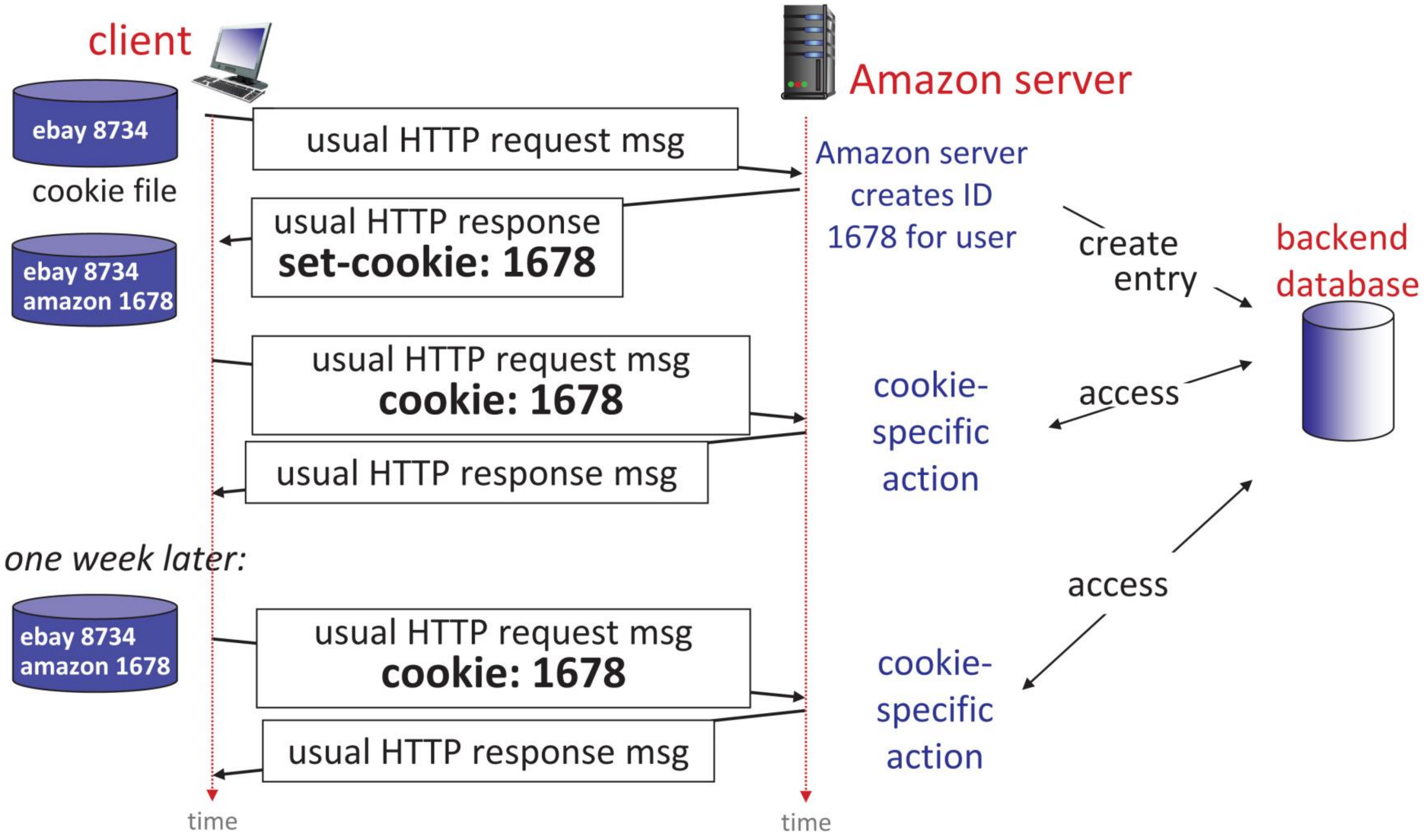
four components:

1. cookie header line of HTTP **response** message
2. cookie header line in next HTTP **request** message
3. cookie file kept on user's host, managed by user's browser
4. back-end database at Web site

Example:

- Susan uses browser on laptop, visits specific e-commerce site for first time
- when initial HTTP requests arrives at site, site creates:
  - unique ID (aka "cookie")
  - entry in backend database for ID
- subsequent HTTP requests from Susan to this site will contain cookie ID value, allowing site to "identify" Susan

Maintaining User/Server State: Cookies (3 of 3)



HTTP Cookies: Comments

What cookies can be used for:

- authorization
- shopping carts
- recommendations
- user session state (Web e-mail)

Challenge: How to keep state?

- **at protocol endpoints:** maintain state at sender/receiver over multiple transactions
- **in messages:** cookies in HTTP messages carry state

aside

cookies and privacy:

- cookies permit sites to **learn** a lot about you on their site.
- third party persistent cookies (tracking cookies) allow common identity (cookie value) to be tracked across multiple web sites

Cookies: Tracking a User's Browsing Behavior (3 of 3)

Cookies can be used to:

- track user behavior on a given website (first party cookies)
- track user behavior across multiple websites (third party cookies) without user ever choosing to visit tracker site (!)
- tracking may be **invisible** to user:
  - rather than displayed ad triggering HTTP GET to tracker, could be an invisible link

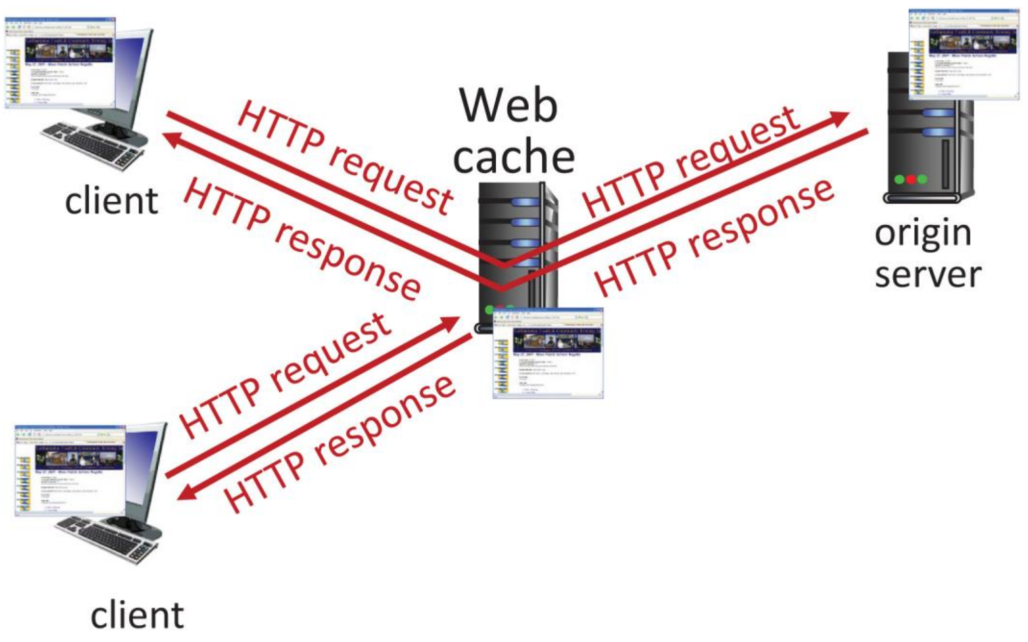
third party tracking via cookies:

- disabled by default in Firefox, Safari browsers
- to be disabled in Chrome browser in 2023

Web Caches

Goal: satisfy client requests without involving origin server

- user configures browser to point to a (local) **Web cache**
- browser sends all HTTP requests to cache
  - **if** object in cache: cache returns object to client
  - **else** cache requests object from origin server, caches received object, then returns object to client



Web Caches (Aka Proxy Servers)

- Web cache acts as both client and server
  - server for original requesting client
  - client to origin server
- server tells cache about object's allowable caching in response header:

```
Cache-Control: max-age=<seconds>

Cache-Control: no-cache
```

Why Web caching?

- reduce response time for client request
  - cache is closer to client
- reduce traffic on an institution's access link
- Internet is dense with caches
  - enables "poor" content providers to more effectively deliver content

Caching Example

Scenario:

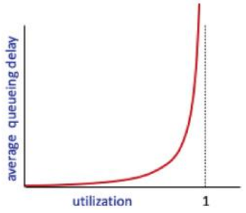
- access link rate: 1.54 Mbps
- RTT from institutional router to server: 2 sec
- web object size: 100K bits
- average request rate from browsers to origin servers: 15/sec
    - avg data rate to browsers: 1.50 Mbps

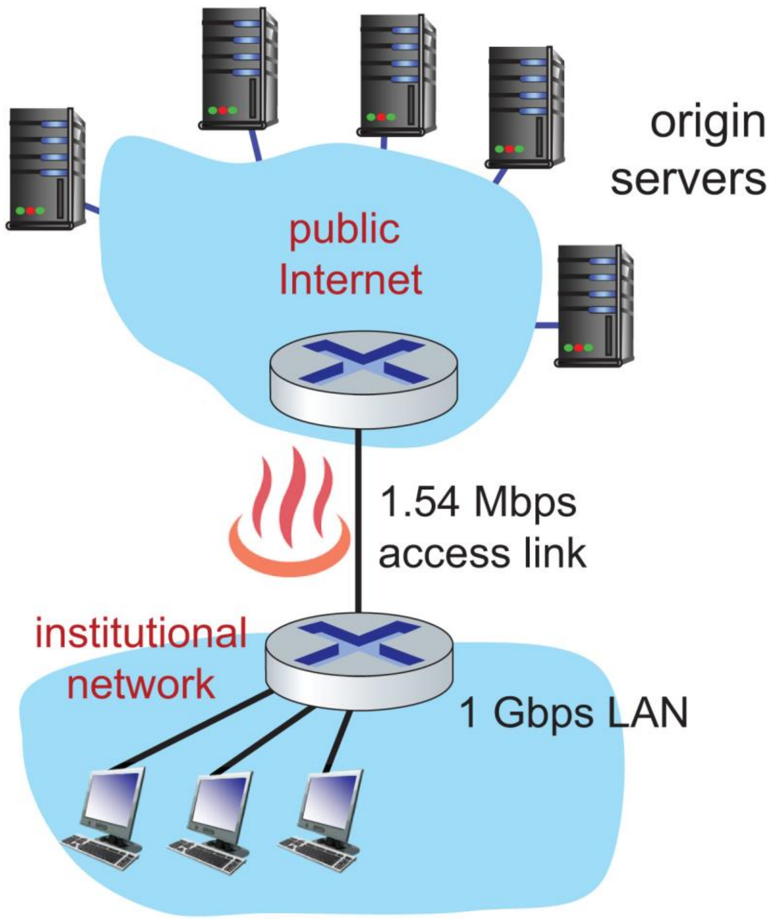
Performance:

- access link utilization = .97
- LAN utilization: .0015
- end-end delay = Internet delay + access link delay + LAN delay

  = 2 sec + minutes + usecs

problem: large queueing delays at high utilization!





Option 1: Buy a Faster Access Link

Scenario:

154 Mbps

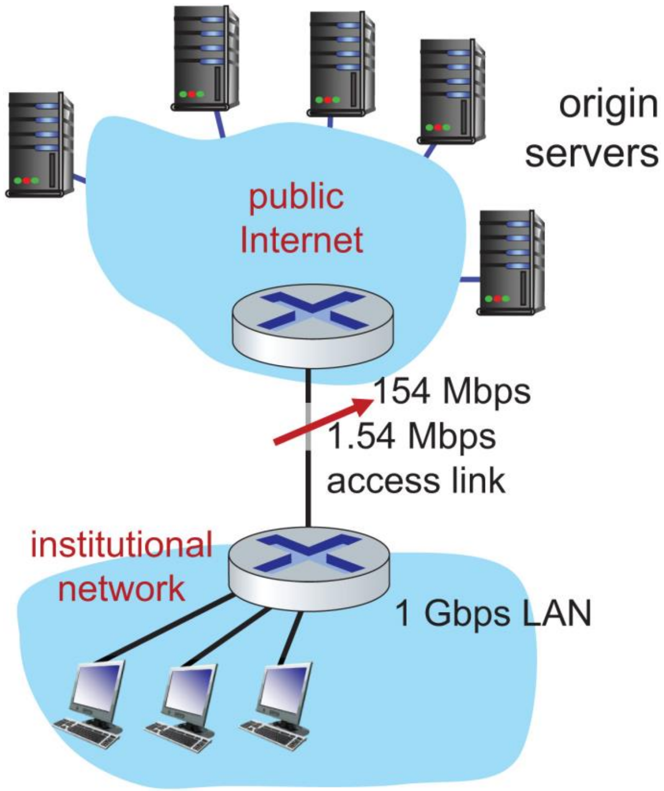
- access link rate: ~~1.54 Mbps~~
- RTT from institutional router to server: 2 sec
- web object size: 100K bits
- average request rate from browsers to origin servers: 15/sec
  - avg data rate to browsers: 1.50 Mbps

Performance:

- access link utilization = ~~.97~~ → .0097
- LAN utilization: .0015
- end-end delay = Internet delay + access link delay + LAN delay
  = 2 sec + ~~minutes~~ + usecs

msecs

Cost: faster access link (expensive!)



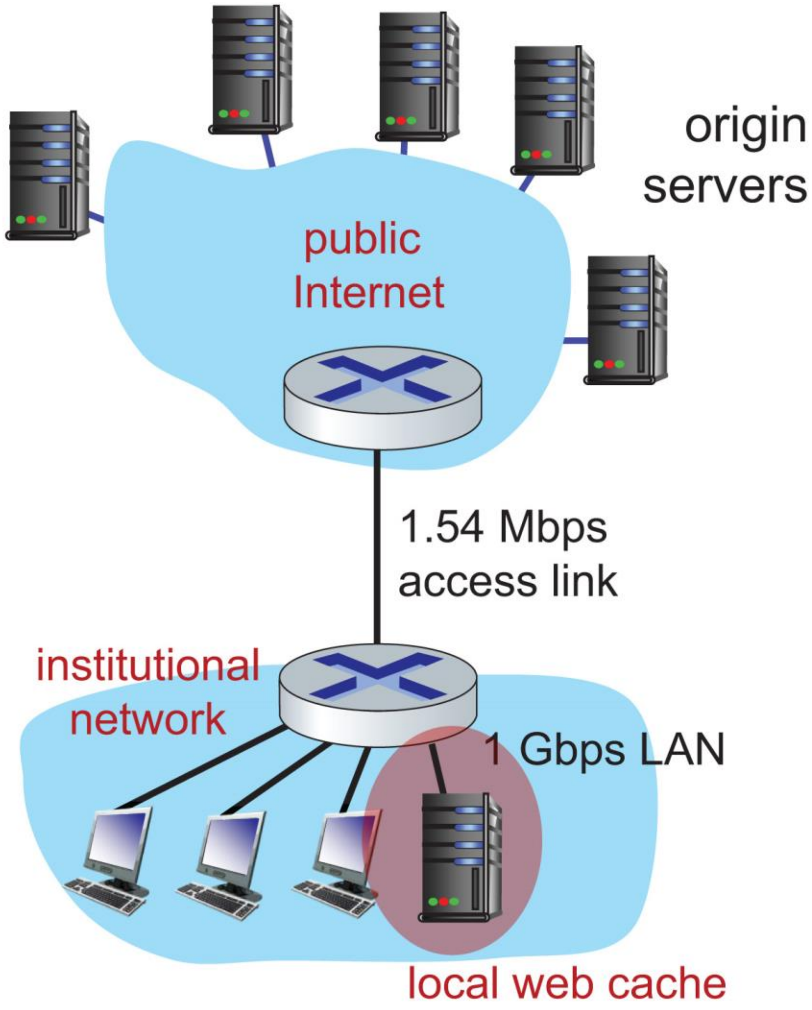
Option 2: Install a Web Cache

Scenario:
- access link rate: 1.54 Mbps
- RTT from institutional router to server: 2 sec
- web object size: 100K bits
- average request rate from browsers to origin servers: 15/sec
  - avg data rate to browsers: 1.50 Mbps

Cost: web cache (cheap!)

Performance:
- LAN utilization: .?
- access link utilization = ?
- average end-end delay = ?

How to compute link utilization, delay?



Calculating Access Link Utilization, end-end Delay With Cache:

suppose cache hit rate is 0.4:

- 40% requests served by cache, with low (msec) delay
- 60% requests satisfied at origin
  - rate to browsers over access link

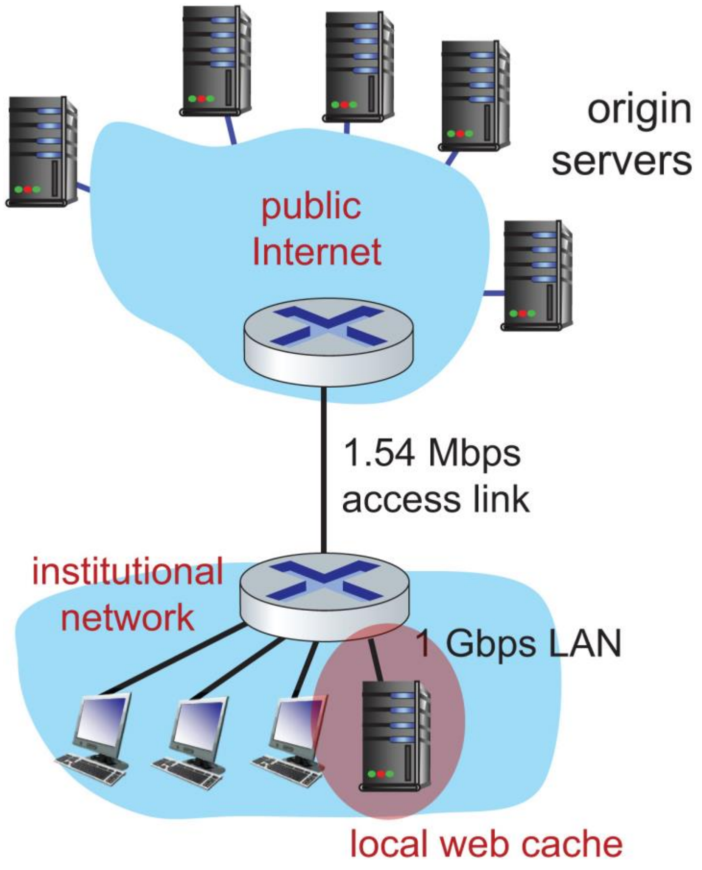
    $= 0.6 * 1.50 \text{ Mbps} = .9 \text{ Mbps}$
  - access link utilization $= 0.9 / 1.54 = .58$ means

    low (msec) queueing delay at access link
- average end-end delay:

  $= 0.6 * (\text{delay from origin servers})$

  $+ 0.4 * (\text{delay when satisfied at cache})$

  $= 0.6 \, (2.01) + 0.4 \, (\sim \text{msecs}) = \sim 1.2 \text{ secs}$



lower average end-end delay than with 154 Mbps link (and cheaper too!)

Browser Caching: Conditional GET

Goal: don't send object if browser has up-to-date cached version

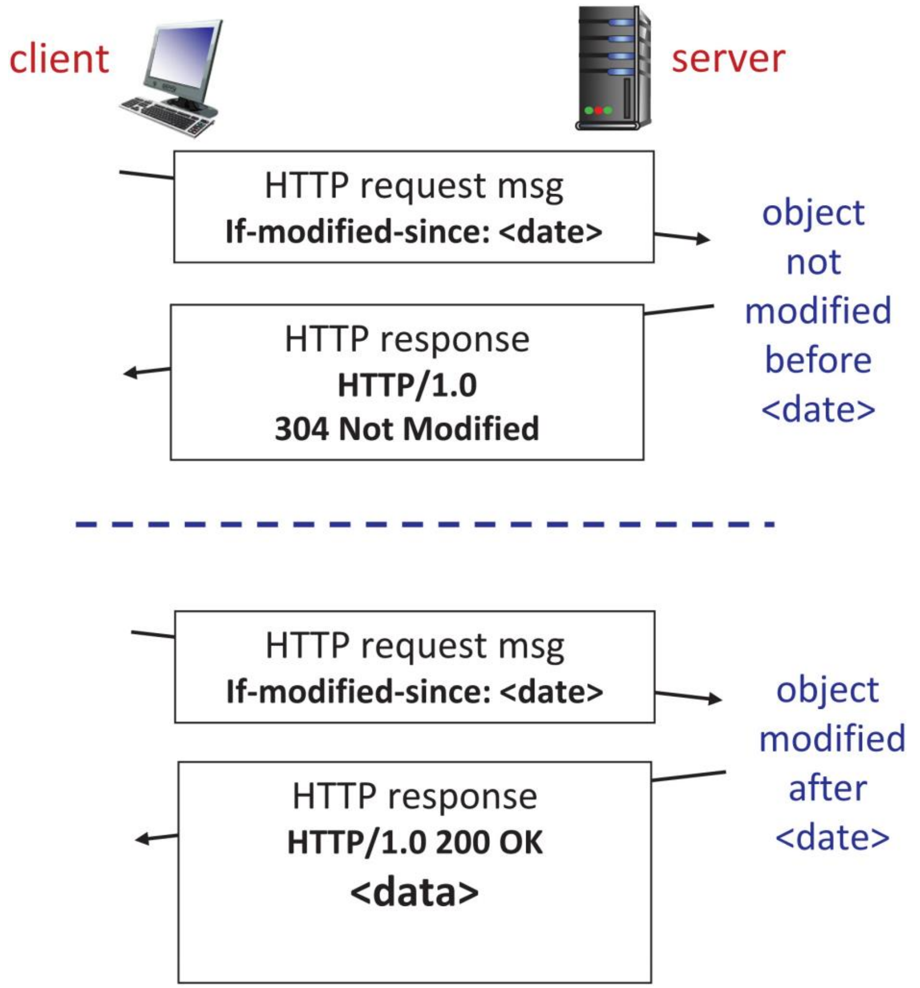
- no object transmission delay (or use of network resources)

- client: specify date of browser-cached copy in HTTP request

  If-modified-since: <date>

- server: response contains no object if browser-cached copy is up-to-date:

  HTTP/1.0 304 Not Modified



HTTP/2 (1 of 2)

Key goal: decreased delay in multi-object HTTP requests

HTTP1.1: introduced multiple, pipelined GETs over single TCP connection

- server responds **in-order** (FCFS: first-come-first-served scheduling) to GET requests
- with FCFS, small object may have to wait for transmission (head-of-line (HOL) blocking) behind large object(s)
- loss recovery (retransmitting lost TCP segments) stalls object transmission

HTTP/2 (2 of 2)

Key goal: decreased delay in multi-object HTTP requests

HTTP/2: [RFC 7540, 2015] increased flexibility at **server** in sending objects to client:

- methods, status codes, most header fields unchanged from HTTP 1.1
- transmission order of requested objects based on client-specified object priority (not necessarily FCFS)
- **push** unrequested objects to client
- divide objects into frames, schedule frames to mitigate HOL blocking

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